

1 (a) (i) $Q = It$ (allow any subject for the equation) B1 [1]

(ii) $\frac{I}{t}$ B1
B1 [2]

(allow 1 mark only if all three quoted)

(b) (i) base unit of I is A
base unit of n is m^{-3} (not $/\text{m}^{-3}$)
base unit of S is m^2
base unit of q is A s (not C)
base unit of v is m s^{-1}
(–1 for each error or omission) B3 [3]

(ii) $A = \text{m}^{-3} \text{m}^2 \text{A s} (\text{m s}^{-1})^k$ M1
e.g. for m: $0 = -3 + 2 + k$
 $k = 1$ A1 [2]

2 (a) (i) $v^2 = 2as$